

NayePankh Volunteer Analytics & Prediction System

Project Report

Submitted By

Yogiraj Gajendra Pawar

B.Tech Computer Science Engineering

D. Y. Patil University, School of Engineering and Technology, Pune

1. Introduction

NayePankh Foundation is a non-profit organization that focuses on education, social welfare, volunteer engagement, and community development. Managing volunteers efficiently is essential for ensuring successful execution of events and campaigns.

This project, "NayePankh Volunteer Analytics & Prediction System," demonstrates how Data Analytics, Machine Learning, and Artificial Intelligence can be used to improve volunteer management. The system analyzes volunteer data, identifies participation trends, predicts future volunteer engagement, and provides intelligent recommendations for volunteer retention.

2. Objectives

The main objectives of this project are:

- To analyze volunteer participation data.
- To identify patterns and trends in volunteer engagement.
- To visualize volunteer activities using charts and graphs.
- To predict future volunteer participation using Machine Learning.
- To provide AI-based recommendations for volunteer management.
- To demonstrate how data-driven decision-making can support NGO operations.

3. Technologies Used

The following technologies and tools were used:

- Python
 - Pandas
 - NumPy
 - Matplotlib
 - Seaborn
 - Scikit-Learn
 - Google Colab
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4. Dataset Description

A volunteer dataset was created containing information related to volunteer activities.

Dataset Features

Attribute	Description
Volunteer_ID	Unique volunteer identifier
Age	Age of volunteer
City	Volunteer's city
Event	Event participated in
Hours_Contributed	Total volunteer hours
Previous_Participation	Number of previous events attended
Training_Attended	Whether volunteer attended training
Participated_Next_Event	Future participation status

The dataset contains volunteer records from multiple cities and event categories.

5. Data Analytics

Data Analytics was performed to understand volunteer behavior and participation patterns.

Analyses Performed

Volunteer Distribution by City

This analysis identifies cities with the highest volunteer participation.

Event Participation Analysis

Different event categories were analyzed to determine which activities attract the highest number of volunteers.

Average Hours Contributed

Volunteer contribution hours were compared across various events.

Age Distribution

The age profile of volunteers was analyzed to understand the target demographic.

Training Participation

The relationship between training attendance and volunteer engagement was examined.

6. Data Visualization

Several visualizations were created to present the data effectively.

Visualizations Included

- Bar Chart: Volunteers by City
- Count Plot: Event Participation
- Histogram: Age Distribution
- Pie Chart: Training Attendance
- Heatmap: Confusion Matrix

These visualizations help stakeholders understand volunteer engagement trends quickly and effectively.

7. Machine Learning Model

A Machine Learning model was developed to predict whether a volunteer is likely to participate in future events.

Algorithm Used

Logistic Regression

Input Features

- Age
- Hours Contributed
- Previous Participation
- Training Attended

Output

- Participate in Next Event (Yes/No)

Model Evaluation

The model was evaluated using:

- Accuracy Score
- Confusion Matrix

The prediction model can help the foundation identify highly engaged volunteers and improve retention strategies.

8. Artificial Intelligence Component

An AI-based recommendation system was developed to classify volunteers according to their engagement levels.

Categories

Highly Engaged Volunteer

Volunteers with:

- High contribution hours
- Multiple previous participations

Needs Encouragement

Volunteers requiring additional engagement and motivation.

Benefits

- Improves volunteer retention
 - Helps prioritize outreach efforts
 - Supports personalized volunteer engagement strategies
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9. Key Findings

The analysis revealed several important insights:

Finding 1

Volunteer participation varies across different cities.

Finding 2

Certain event types attract more volunteers than others.

Finding 3

Volunteers who attend training programs are more likely to participate in future events.

Finding 4

Previous participation history strongly influences future engagement.

Finding 5

Machine Learning can effectively predict volunteer participation patterns.

10. Benefits for NayePankh Foundation

The proposed system offers several benefits:

- Better volunteer management
 - Improved event planning
 - Increased volunteer retention
 - Data-driven decision making
 - Early identification of inactive volunteers
 - Enhanced resource allocation
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11. Future Enhancements

The project can be extended further by:

- Creating an interactive Power BI dashboard
 - Using larger real-world datasets
 - Implementing advanced Machine Learning models
 - Developing a web-based volunteer management portal
 - Integrating an AI-powered chatbot
 - Automating report generation
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12. Conclusion

The NayePankh Volunteer Analytics & Prediction System demonstrates how modern technologies such as Data Analytics, Machine Learning, and Artificial Intelligence can support NGO operations.

By analyzing volunteer data, predicting future participation, and providing intelligent recommendations, the system helps improve volunteer engagement and operational efficiency. The project highlights the importance of data-driven approaches in strengthening social impact initiatives and supporting the long-term growth of NayePankh Foundation.

References

1. Python Documentation
2. Pandas Documentation
3. Scikit-Learn Documentation
4. Matplotlib Documentation
5. Seaborn Documentation
6. NayePankh Foundation Website